

Research Agenda: From Black-Box to White-Box Understanding of the Autoresearch Loop

What I want to learn. I want to learn how a compound, LLM-driven research system can be made interpretable in a precise sense: not by inspecting its weights, but by establishing an explicit and testable causal link between its instruction file (program.md) and its observable research behavior. The underlying methodological question is how a compositional instruction file can be decomposed into a small set of interpretable components, and how the causal contribution of each component to system behavior can be measured. This is structurally the same question the AutoML literature answers for hyperparameters, and I intend to reuse that methodology rather than invent a new one.

What I want to achieve. Concretely, I want to determine how variations in program.md determine the behavior of the autoresearch loop when it is tasked with researching random forests and gradient boosting. Restricting the domain keeps the space of possible hypotheses, experiments, and outputs small enough to analyze exhaustively. Behavior is defined at two levels: the process (which hypotheses are formed, which experiments are designed, in what order, with what stopping criterion) and the product (code, results, final write-up). The white-box goal is reached when I can state, for a given change to program.md, which behavioral feature it will change and by how much, not merely that the outputs differ.

Research agenda.

1. *Human baseline.* Perform the same random forest / gradient boosting research task myself under an instruction equivalent to program.md, using think-aloud protocol analysis, to obtain a reference trace of research actions (hypothesis formation, experiment design, evaluation, iteration, stopping).
2. *Program.md as a configuration space.* Decompose program.md into a finite set of interpretable components (e.g. evaluation metric specified or not, exploration breadth, stopping criterion, degree of ordering/emphasis) — treated the same way a hyperparameter vector is treated in AutoML.
3. *Ablation instead of random sampling.* Vary these components using a factorial/ablation design, as is standard in AutoML hyperparameter-importance studies (e.g. functional ANOVA, SMAC-style configuration analysis), and run the loop repeatedly per configuration to capture stochastic variance, rather than sampling program.md variants unsystematically.
4. *Behavioral encoding.* Encode each run's process and product as a small set of quantitative features (algorithmic variants explored, search breadth/depth, distance from the human baseline trace, novelty/quality of findings), extending the semantic/behavioral distance measures already developed.
5. *Surrogate model (the white box).* Fit an interpretable model (functional ANOVA, shallow decision tree, or linear model with interaction terms) mapping program.md features to behavioral features. Interpreting this surrogate — which components matter, and how much — is the white box.
6. *Causal validation.* Use the surrogate to predict a program.md configuration that should produce a specific behavior, run the loop, and check whether the prediction holds. This step is what distinguishes a white box from a merely correlational (black-box) account.